SpaceX Falcon 9 First Stage Landing Prediction

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

IBM Data Science Professional Certificate Capstone Project

Satish Kumar Ram

4 August 2026

Outline

- Executive Summary
- Introduction
- Data Collection & Data Wrangling Methodology
- EDA & Data Visualization Methodology
- EDA with Visualization
- EDA with SQL Queries
- Folium Map
- Plotly Dash Dashboard
- Predictive Analysis (Classification)
- Predictive Analysis Results
- Conclusion
- Appendix

Introduction

- **Project Background and Context:**

- SpaceX has significantly reduced the cost of space missions by successfully recovering and reusing Falcon 9 first-stage boosters. Traditional rocket launches are expensive because rockets are usually discarded after launch. By predicting whether the Falcon 9 first stage will land successfully, organizations can estimate mission costs, improve planning, and support data-driven decision-making. This project uses historical SpaceX launch data to analyze landing outcomes and build machine learning models for landing prediction.

- **Project Objective:**

- The project aims to identify the factors influencing Falcon 9 first-stage landing success and develop a machine learning model to predict landing outcomes.
- Collect Falcon 9 launch data using the SpaceX REST API and **Wikipedia Web Scraping**.
- Perform **data wrangling, SQL analysis, and exploratory data analysis (EDA).**
- Build interactive visualizations using SQL, Folium, and Plotly Dash.
- Develop, evaluate, and compare machine learning classification models to predict Falcon 9 first-stage landing success.

Executive Summary

- **Objective**
  - Predict Falcon 9 first-stage landing success using historical launch data and machine learning.
- **Summary of Methodology**
  - Data Collection through **SpaceX REST API.**
  - Data Collection using **Wikipedia Web Scraping.**
  - Data Wrangling and Feature Engineering.
  - Exploratory Data Analysis (EDA) using SQL Queries, Python Visualizations (Matplotlib & Seaborn), and Plotly Dash.
  - Interactive Visual Analytics using **Folium** and **Plotly Dash.**
  - Predictive Analysis using **Classification Models.**
- **Summary of Results**
  - Landing success rate improved significantly over time.
  - Payload mass, orbit type, and launch site influenced landing success.
  - SQL and visualizations revealed important launch patterns.
  - Interactive dashboards provided deeper insights into launch data.
  - The best classification model achieved the highest prediction accuracy.

Methodology

- **Data Collection Methodology**
- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Collected additional launch records through **Wikipedia Web Scraping**.
- **Data Wrangling**
- Cleaned missing values and standardized the dataset.
- Created the landing success label and prepared features for analysis.
- **Exploratory Data Analysis (EDA)**
- Performed **SQL queries** to analyze launch statistics.
- Created visualizations using Matplotlib and Seaborn.
- **Interactive Visual Analytics**
- Built an interactive **Folium** map to analyze launch site locations.
- Developed an interactive Plotly Dash dashboard with dropdown filters, charts, and data visualizations for launch analysis.
- **Predictive Analysis**
- Built, trained, and evaluated multiple machine learning classification models.
- Compared model performance and selected the best-performing model based on prediction accuracy.

Data Collection using SpaceX API

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Retrieved launch date, payload mass, orbit, launch site, booster version, and mission outcome.
- Converted JSON responses into a structured Pandas DataFrame for analysis.
- **Right side:** SpaceX API notebook screenshot.

```
import requests
import pandas as pd

static_json_url = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json"

response = requests.get(static_json_url)

print(response.status_code)
```

```
200
```

We should see that the request was successfull with the 200 status response code

```
response.status_code
```

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
# Use json_normalize method to convert the json result into a dataframe
data = pd.json_normalize(response.json())
```

Using the dataframe `data` print the first 5 rows

```
# Get the head of the dataframe
data.head(5)
```

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Data Collection using Wikipedia Web Scraping

- Extracted additional Falcon 9 launch records from **Wikipedia** using **BeautifulSoup**.
- Parsed HTML tables and converted them into a Pandas DataFrame.
- Used the scraped data to enrich and validate the dataset.
- Successfully extracted Falcon 9 launch tables using BeautifulSoup for further analysis.

```python
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = requests.get(static_url, headers=headers)
```

Create a `BeautifulSoup` object from the HTML `response`

```python
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content

soup = BeautifulSoup(response.text, "html.parser")
```

```python
print(soup.title.string)
```

```
List of Falcon 9 and Falcon Heavy launches - Wikipedia
```

Print the page title to verify if the `BeautifulSoup` object was created properly

```python
# Use soup.title attribute
print(soup.title)
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

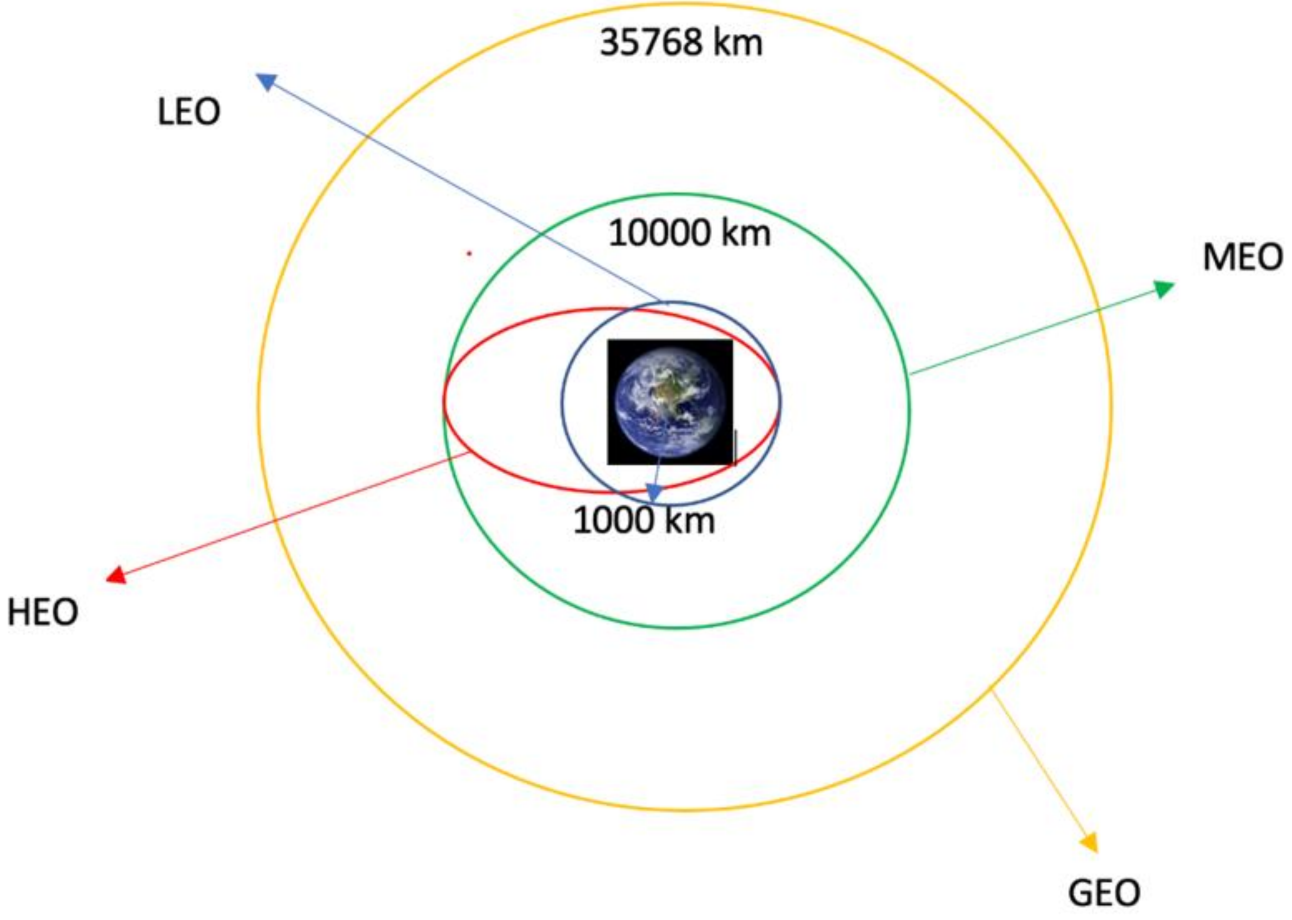
```python
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`

html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

```python
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
print(first_launch_table)
```

Data Wrangling

- Cleaned missing and inconsistent values.
- Selected relevant features for analysis.
- Created the landing success label.
- Prepared the final dataset for EDA and machine learning.

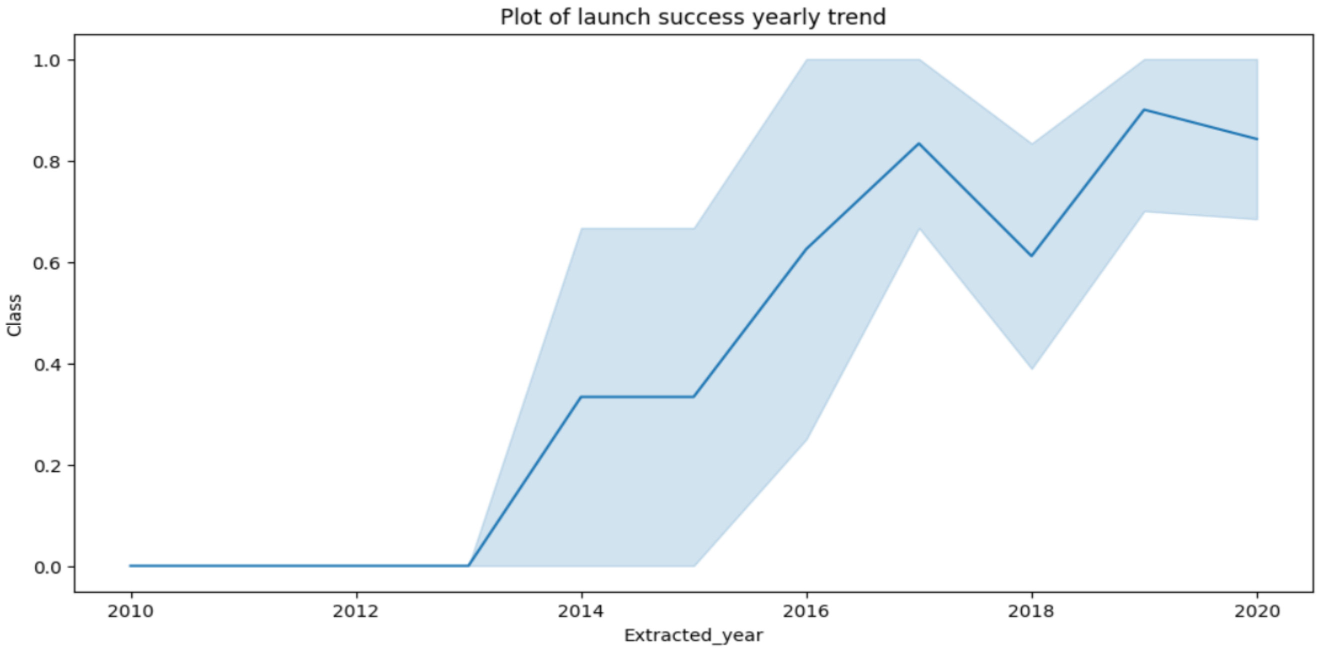


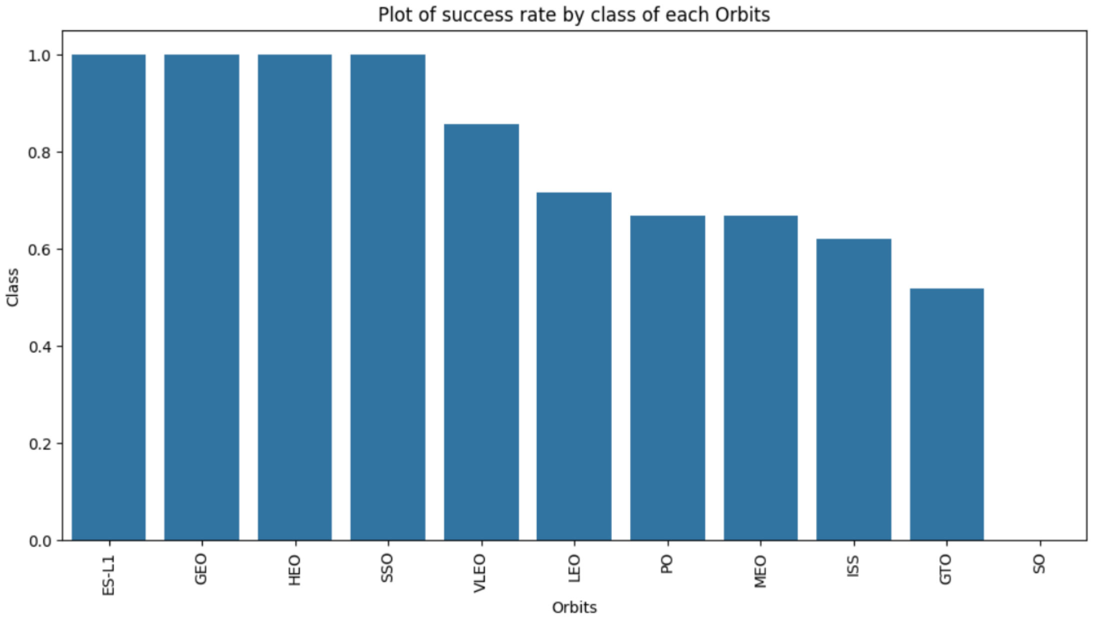
GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

EDA using Visualizations

- Launch success increased over time.
- Success rates differed across orbit classes.
- EDA revealed patterns influencing landing success.





GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

EDA using SQL

- Used SQL queries to summarize launch statistics.
- Analyzed launches by launch site, mission outcome, and booster version.
- Identified trends and generated insights from the data.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

EDA Using SQL Queries (Part 1)

SQL Query Output – Launch Site Analysis

```
%%sql

SELECT Launch_Site,
COUNT(*) AS Total_Launches
FROM SPACEXTABLE
GROUP BY Launch_Site;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site	Total_Launches
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

- **Key Findings:**
- Used SQL to calculate total launches for each Falcon 9 launch site.
- **CCAFS SLC-40** recorded the highest number of launches (**34**).
- **CCAFS LC-40** completed **26** launches during the analyzed period.
- KSC LC-39A completed 25 launches.
- **VAFB SLC-4E** recorded the fewest launches (**16**).
- These SQL results identified the busiest launch sites and supported subsequent visualization and predictive analysis.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

EDA Using SQL Queries (Part 2)

Mission Outcome Analysis (SQL Query Result)

```
%%sql

SELECT Mission_Outcome,
       COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Mission_Outcome
ORDER BY Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Count
Success	30
Failure (in flight)	1

- **Key Findings:**
- SQL was used to summarize mission outcomes.
- **30 missions** were completed successfully.
- Only **1 mission** resulted in failure during the selected period.
- The high success rate indicates improved Falcon 9 reliability.
- SQL aggregation helped validate mission performance before predictive modelling.
- These SQL results supported exploratory data analysis and feature selection for machine learning.

EDA Using SQL Queries (Part 3)

[18]:
```
%%sql

SELECT Booster_Version
FROM SPACEXTABLE
WHERE Payload_Mass__kg_ BETWEEN 4000 AND 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

[18]:

Booster_Version
F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1014
F9 v1.1 B1016
F9 FT B1020
F9 FT B1022
F9 FT B1026
F9 FT B1030
F9 FT B1021.2
F9 FT B1032.1
F9 B4 B1040.1

- **Key Findings:**
- SQL was used to identify booster versions carrying payloads between **4000 kg and 6000 kg**.
- Multiple Falcon 9 booster versions operated within this payload range.
- Medium-weight payload missions were common across several booster versions.
- These findings support payload trend analysis and provide useful insights for predictive modelling.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Interactive Visual Analytics (Folium & Plotly Dash)

- Built interactive **Folium** maps to visualize launch site locations.
- Developed a **Plotly Dash** dashboard with filters and interactive charts.
- Enabled dynamic exploration of launch data.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Predictive Analysis

- Built multiple classification models to predict landing success.
- Compared model performance using evaluation metrics.
- Selected the best-performing model based on prediction accuracy.

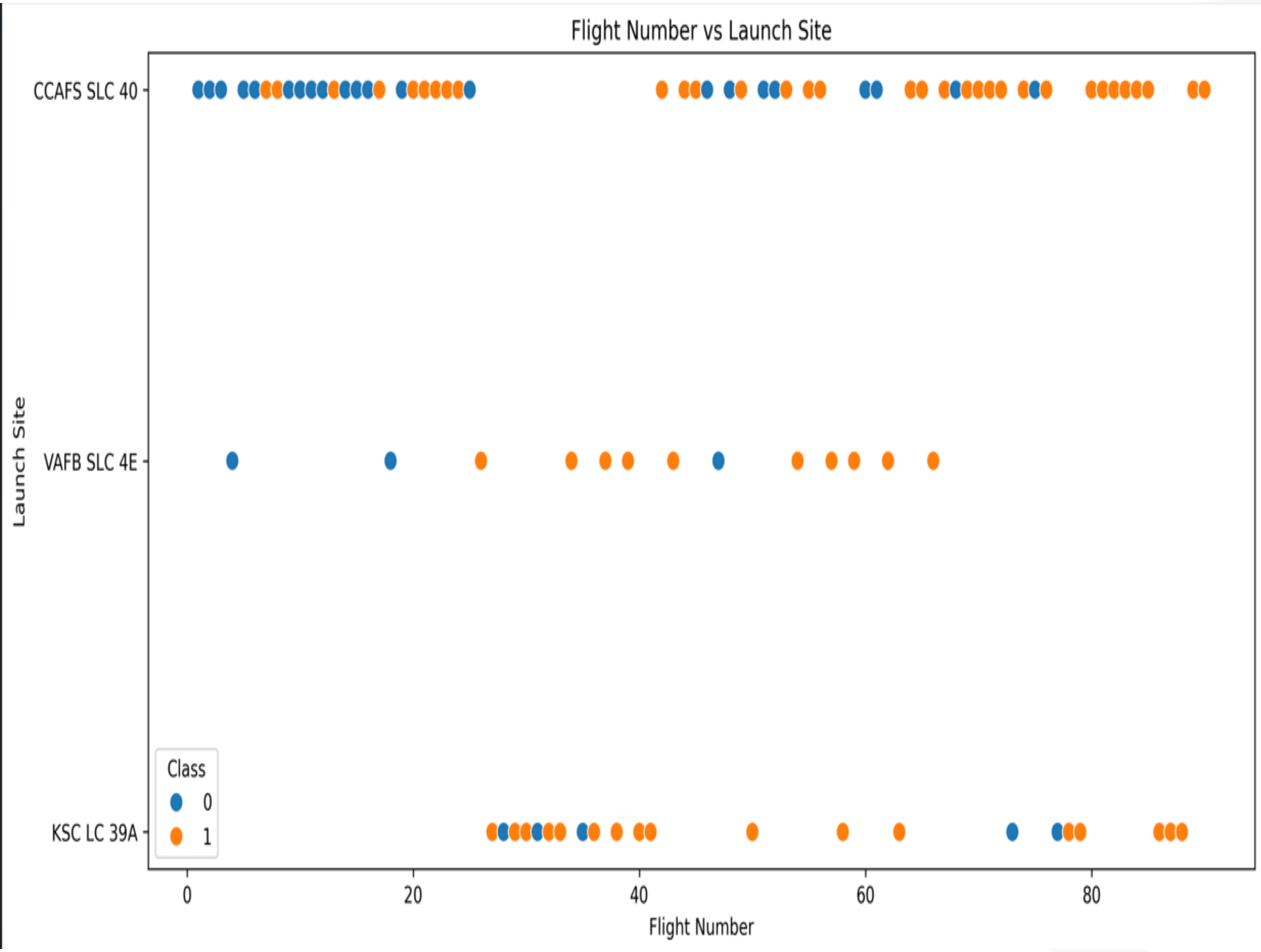
GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Results

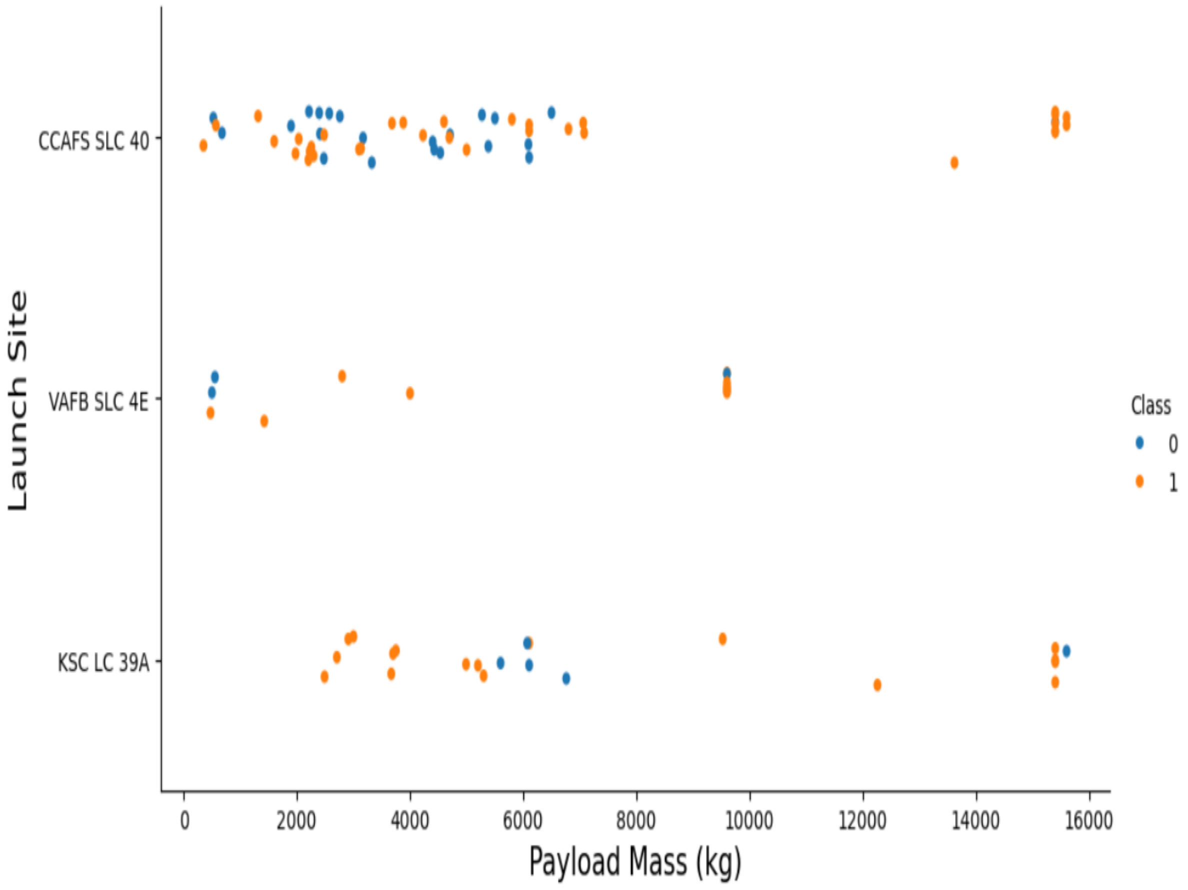
- **Landing Success Trend:** This project integrates data collection, EDA, SQL, interactive dashboards, and machine learning to predict Falcon 9 landing success.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis:** The best classification model achieved the highest prediction accuracy.
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 first-stage landing success and support mission planning.

Flight Number vs. Launch Site



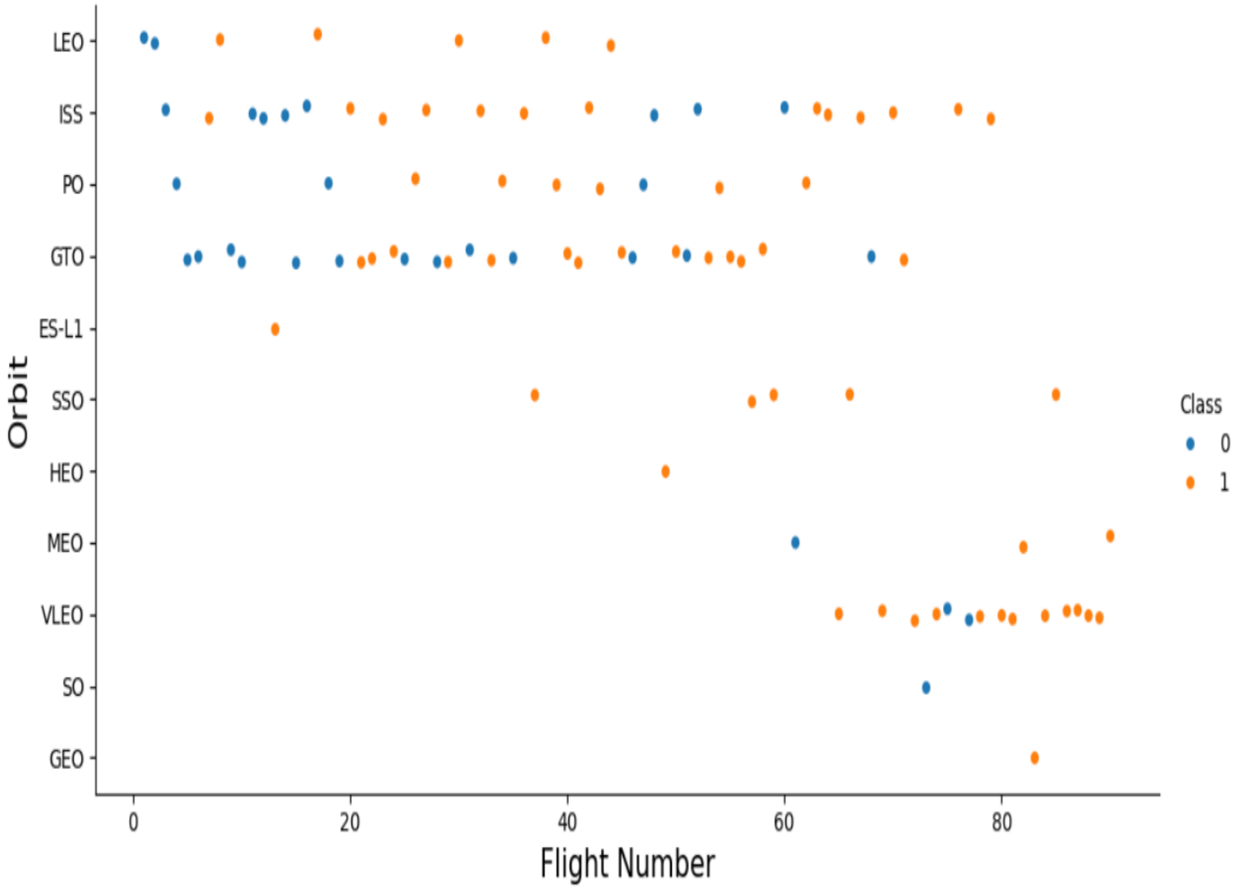
- **Key Findings**
- Early Falcon 9 missions had lower landing success, while later missions showed higher success rates.
- Launch success improved as the **flight number** increased, indicating continuous improvements in booster recovery and operational experience.
- **CCAFS SLC 40** recorded the highest number of Falcon 9 launches, making it the most frequently used launch site.
- **KSC LC 39A** and **VAFB SLC 4E** also demonstrated successful landings, although with fewer launches than CCAFS SLC 40.
- Overall, both launch site and flight number influenced Falcon 9 first-stage landing success.

Payload Mass vs. Launch Site



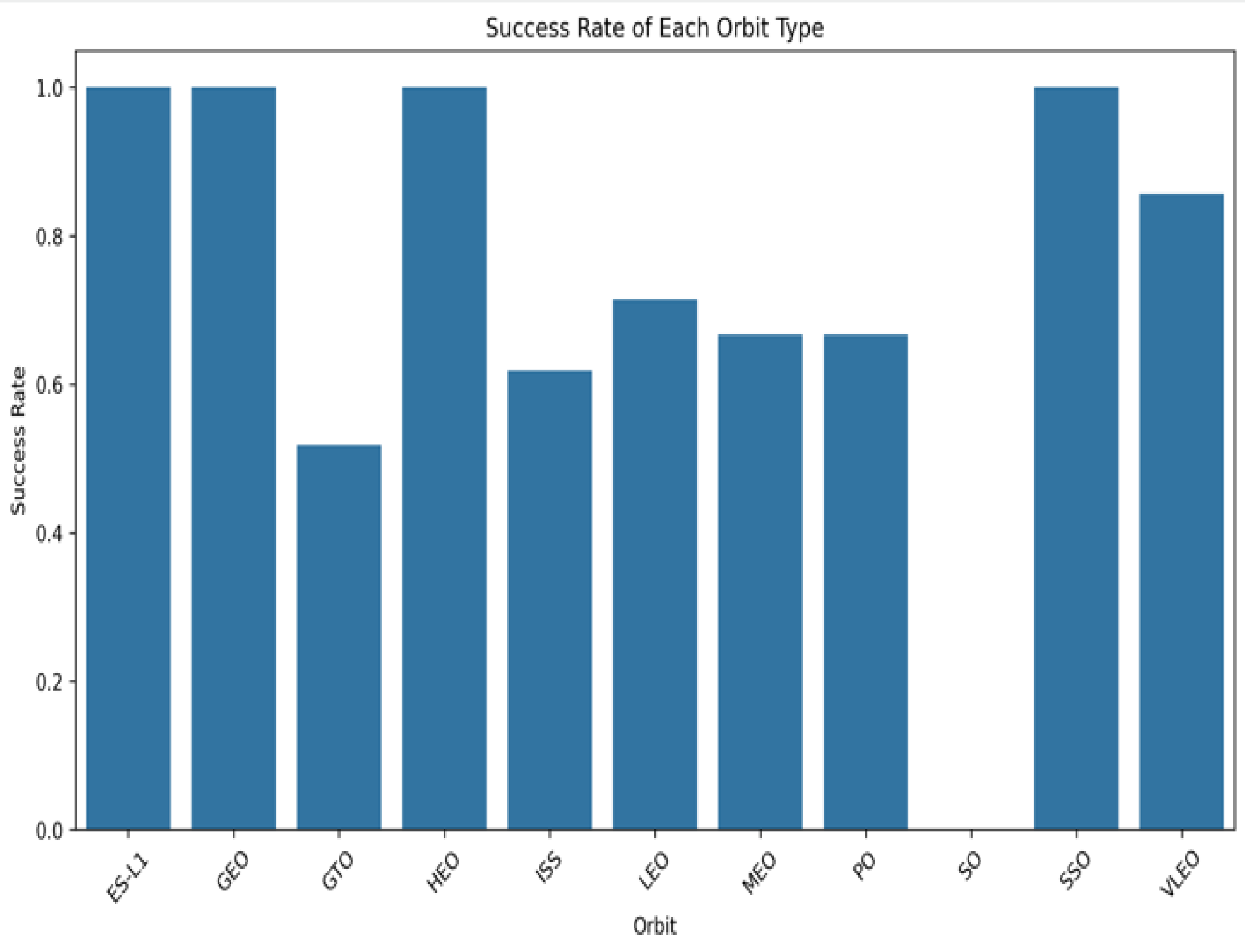
- **Key Findings**
- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, launch site and payload mass together influenced Falcon 9 landing success.

Flight Number vs. Orbit



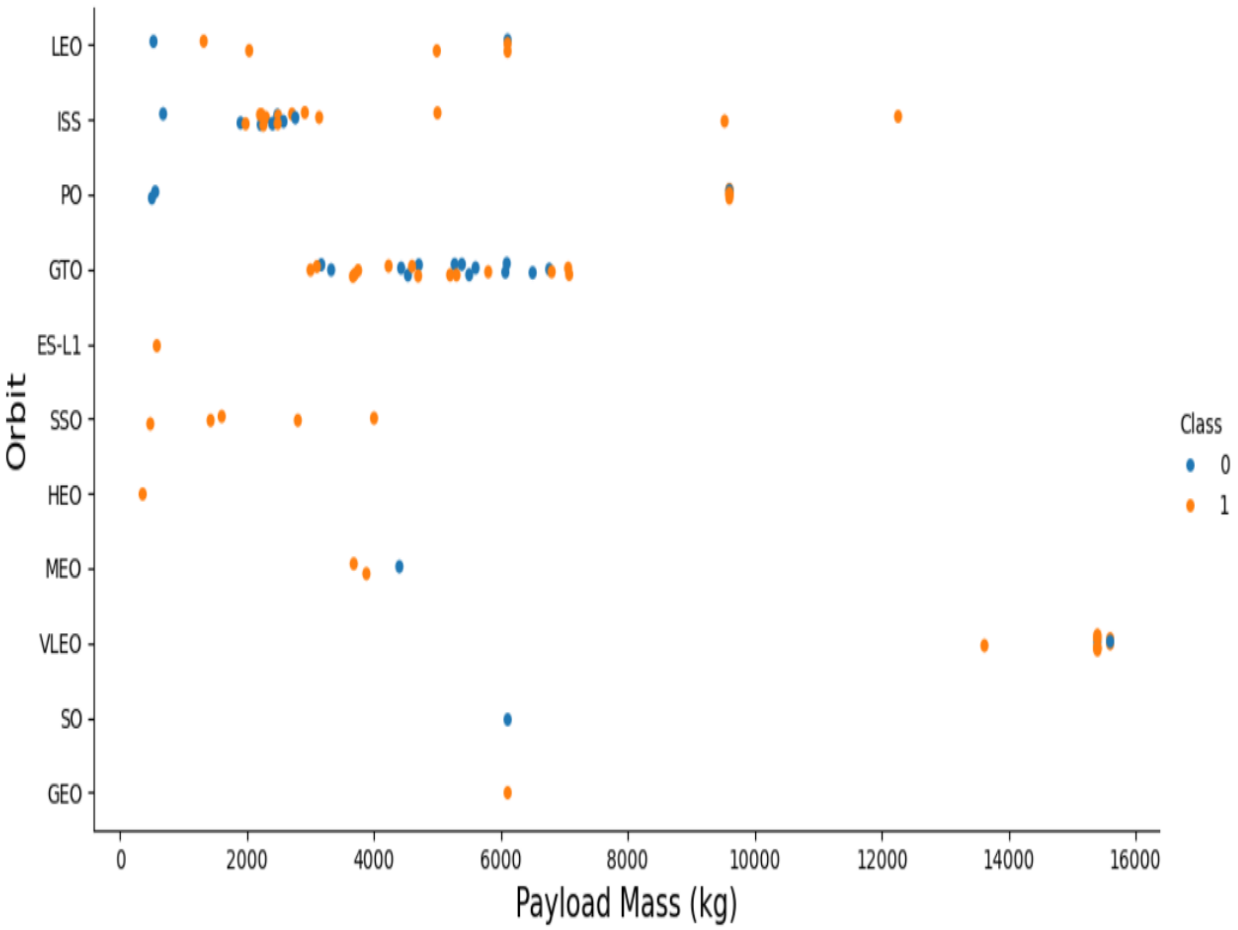
- **Key Findings**
- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Orbit type influenced landing success, while payload mass and launch site also played important roles.

Success Rate by Orbit



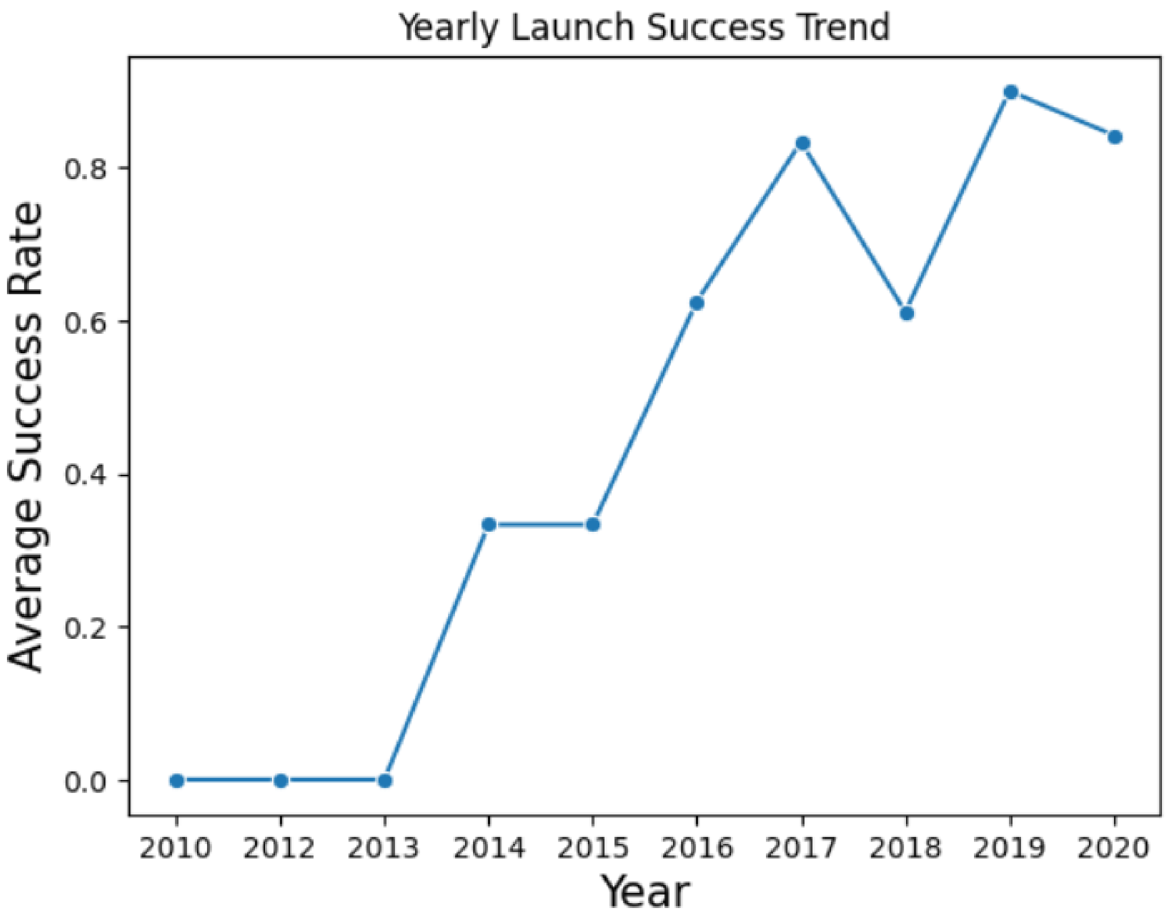
- **Key Findings**
- **ES-L1, GEO, HEO,** and **SSO** achieved a **100% landing success rate**, indicating highly reliable mission outcomes for these orbit types.
- **VLEO** also demonstrated a high landing success rate of approximately **86%**, showing strong booster recovery performance.
- LEO, MEO, ISS, and PO achieved moderate landing success rates, indicating variable mission outcomes across these orbit types.
- **GTO** recorded a comparatively lower landing success rate, while **SO** showed **no successful landings** in the available dataset.
- Overall, orbit type significantly influenced Falcon 9 landing success, although mission characteristics and operational factors also played important roles.

Payload Mass vs. Orbit



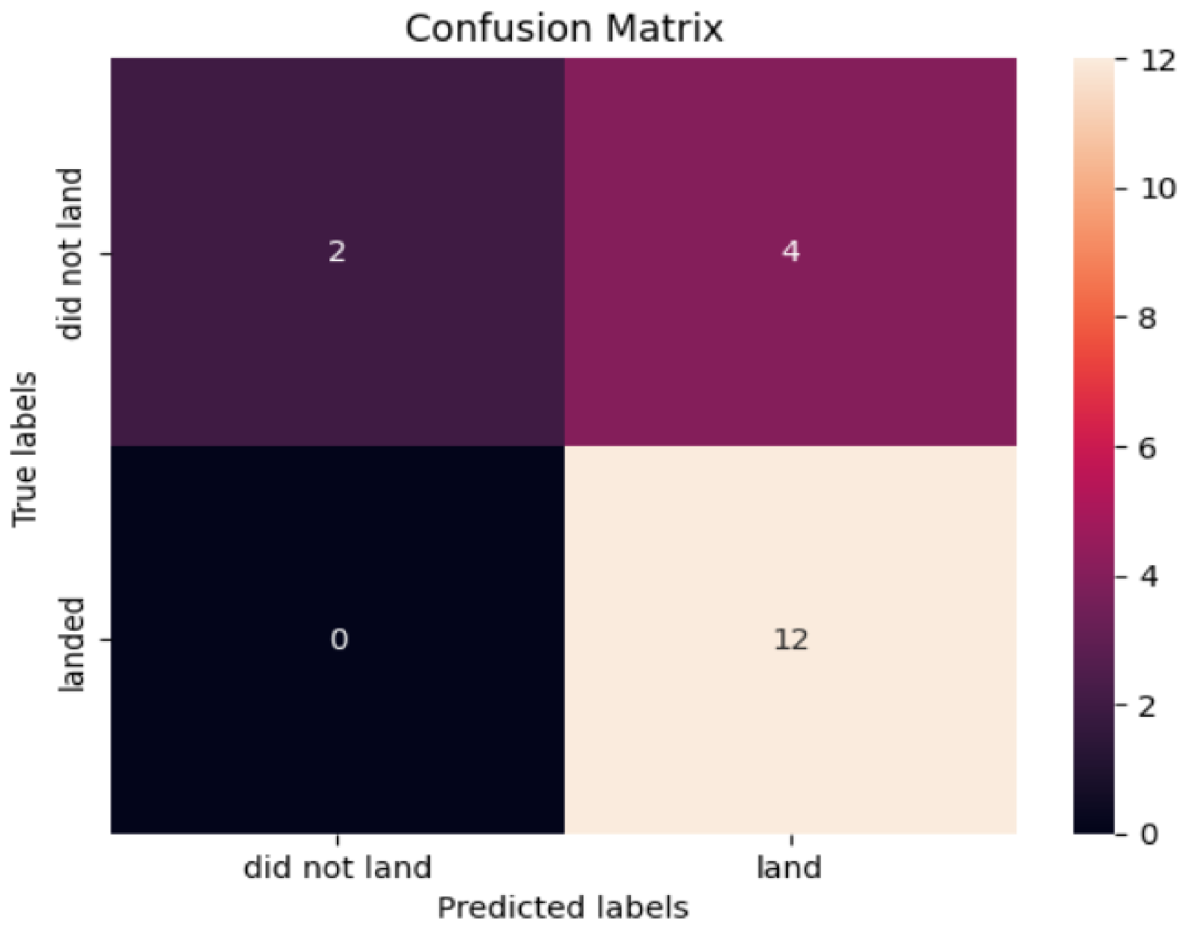
- **Key Findings**
- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low to medium payload masses.
- Payload mass alone did not determine Falcon 9 landing success; orbit type and mission characteristics also influenced landing outcomes.

Yearly Launch Success Trend



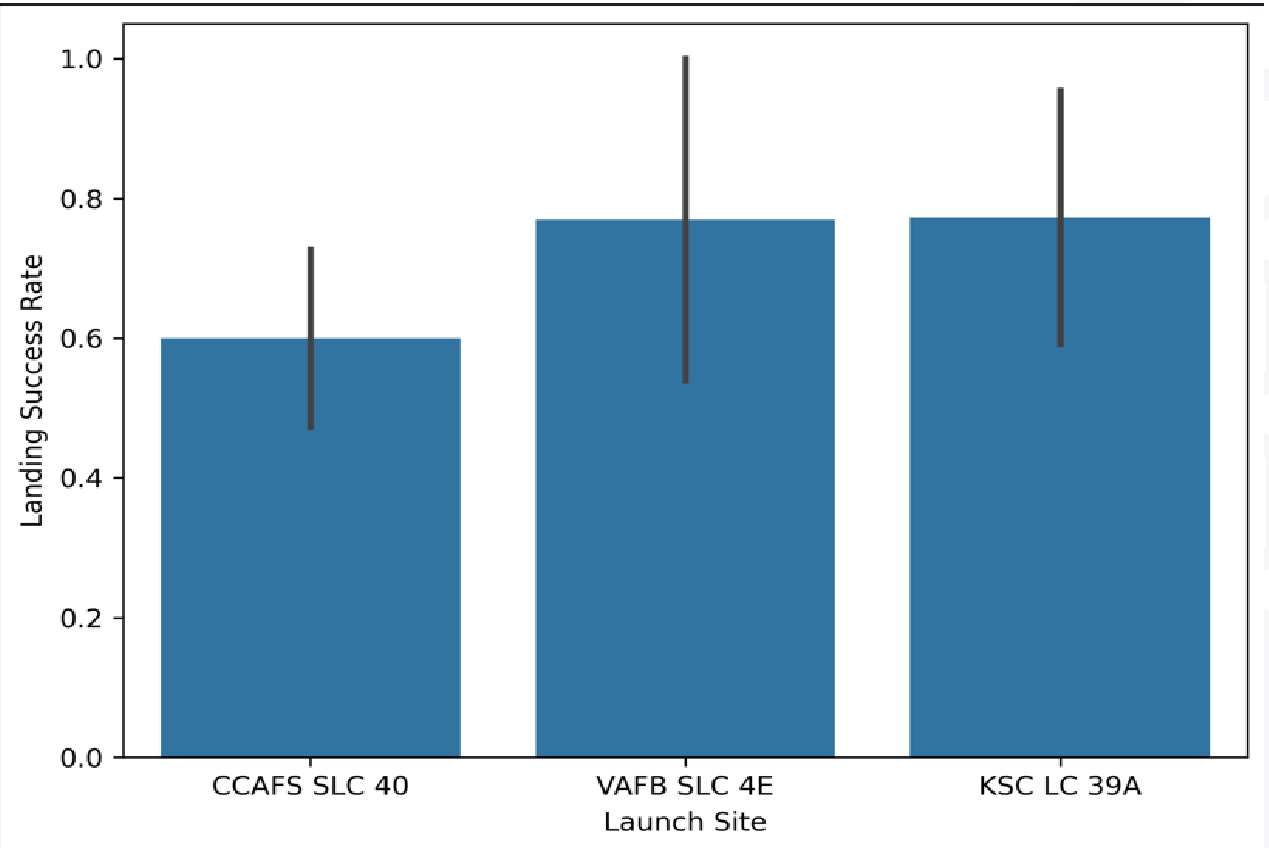
- **Key Findings**
- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onward, average launch success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high launch success rates, demonstrating improved booster recovery and mission reliability.

Model Evaluation – Confusion Matrix



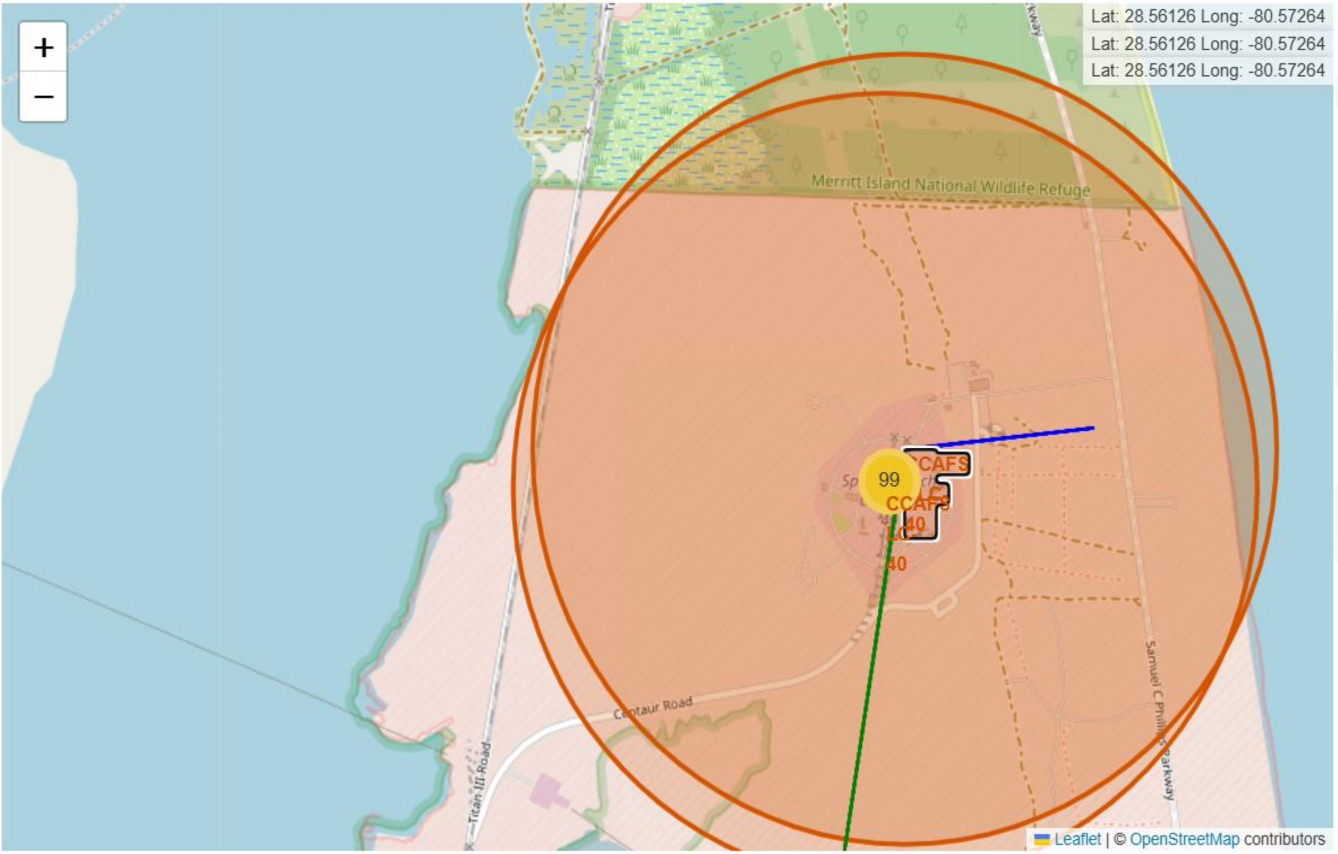
- **Key Findings**
- The model achieved good predictive performance with no false negatives, although a few unsuccessful missions were classified as successful.
- **2 unsuccessful landings** were correctly classified (True Negatives).
- **4 unsuccessful landings** were incorrectly predicted as successful (False Positives).
- The model demonstrated strong performance in identifying successful landings, with no false negatives, although some unsuccessful missions were incorrectly classified as successful.

Landing Success by Launch Site



- **Key Findings**
- **KSC LC 39A** achieved the **highest average landing success rate** among all launch sites.
- **VAFB SLC 4E** also recorded a **high landing success rate**, indicating strong mission performance.
- **CCAFS SLC 40** showed the **lowest average landing success rate** in the dataset.
- Launch site significantly influenced Falcon 9 first-stage landing success.

Launch Site Proximity Analysis (CCAFS)



- **Key Findings:**
- CCAFS launch site is located close to the coastline.
- The coastal location supports safe launch trajectories over the ocean.
- The site is positioned away from major populated areas to improve safety.
- The proximity map highlights CCAFS's strategic coastal location for safe launch operations.

SPACEX FALCON 9 – FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved over time.
- Launch site, payload mass, flight number, and orbit type significantly influenced landing success.
- The best-performing classification model demonstrated strong predictive performance.

Implications

- Machine learning can effectively predict Falcon 9 landing success.
- The predictive model can support mission planning and reduce launch costs.
- Data-driven insights help improve future SpaceX launch decisions.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Results – Visualizations & Dashboard

EDA Visualizations

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Launch Success Trend
- Landing Success by Launch Site

Interactive Dashboard

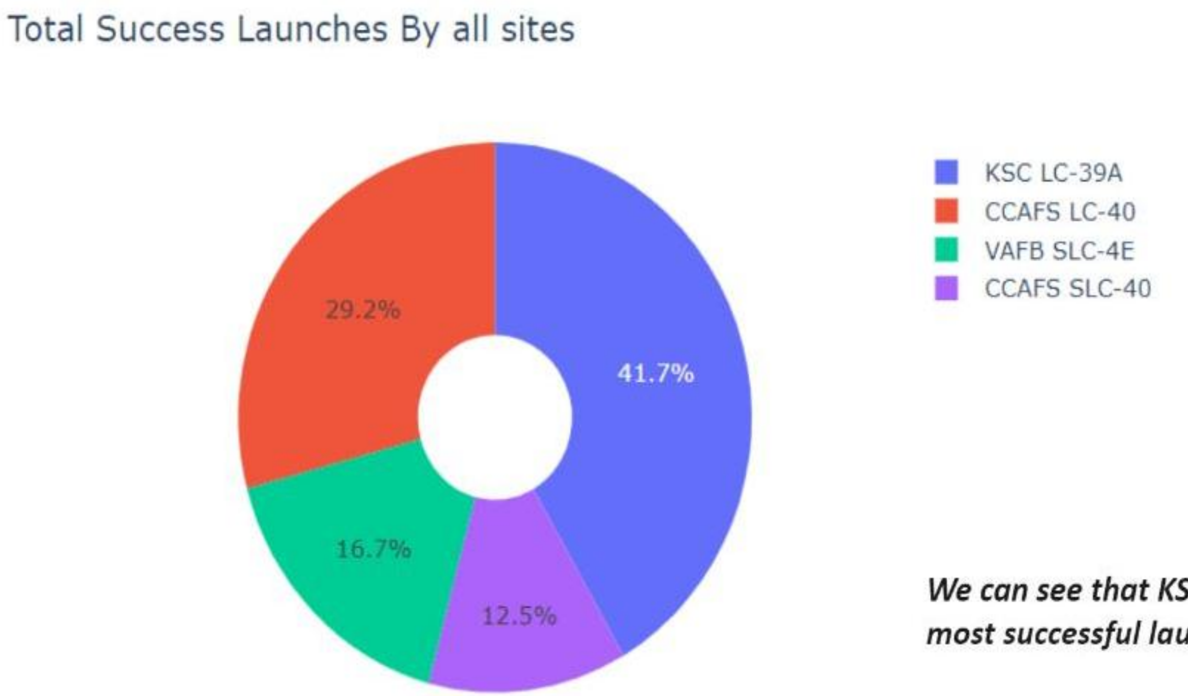
- SQL Query Results
- Folium Launch Site Map
- Plotly Dash Dashboard
- Confusion Matrix (Model Evaluation)

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Interactive Dashboard (Plotly Dash)

Total Success Launches by All Sites (Pie Chart)



We can see that KSC LC-39A had the most successful launches from all the sites

- **Key Findings:**

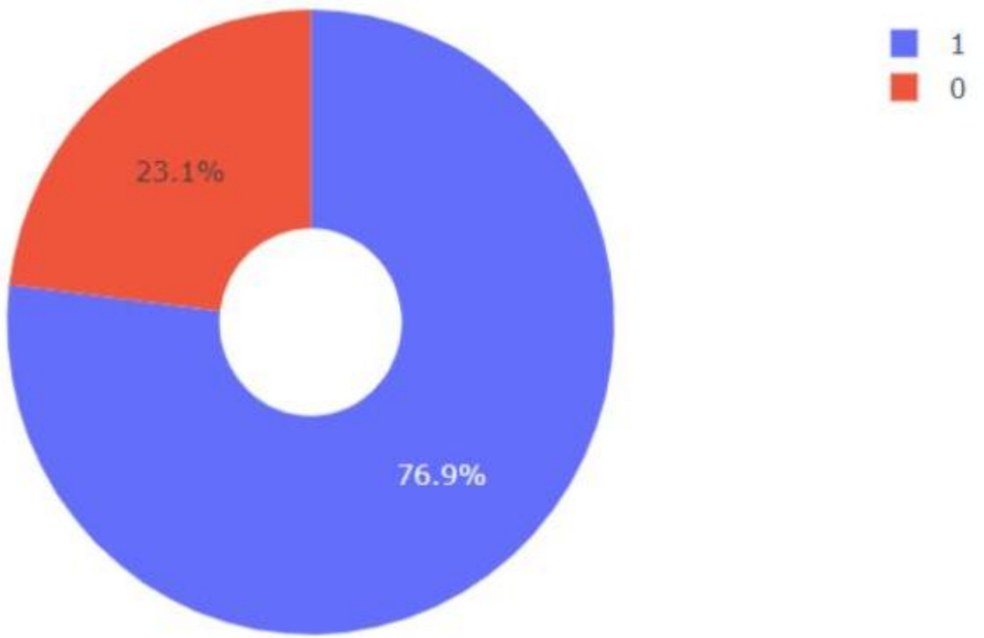
- Interactive dashboard developed using Plotly Dash.
- Displays successful launches across all launch sites.
- KSC LC-39A recorded the highest number of successful launches.
- Figure: Plotly Dash dashboard showing the success distribution across all launch sites.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Launch Site Analysis

KSC LC-39A: Success vs Failure



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

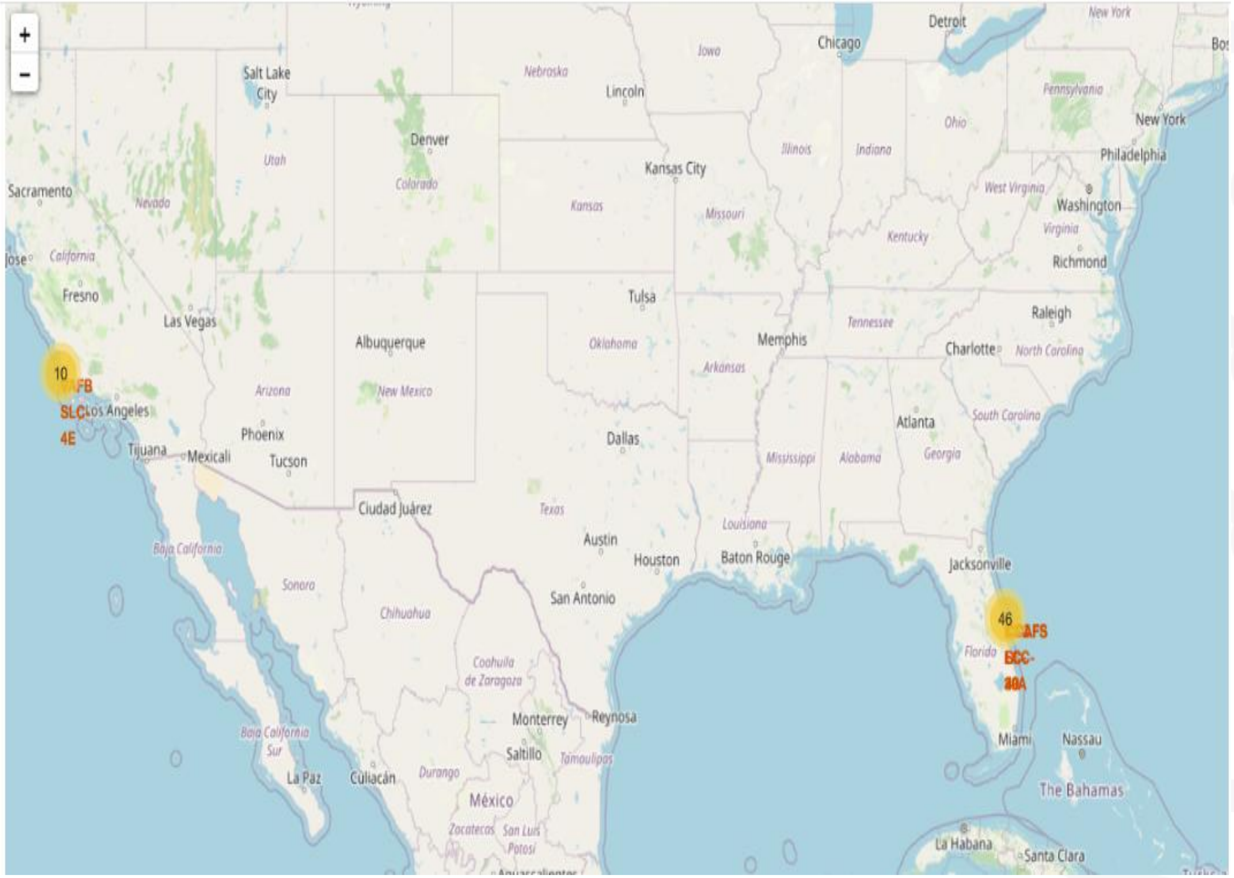
- **Key Findings:**
- Shows the success and failure distribution for **KSC LC-39A**.
- Approximately **76.9%** of launches were successful.
- Helps compare mission outcomes at the selected launch site.
- Demonstrates the effectiveness of Falcon 9 operations at KSC LC-39A.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

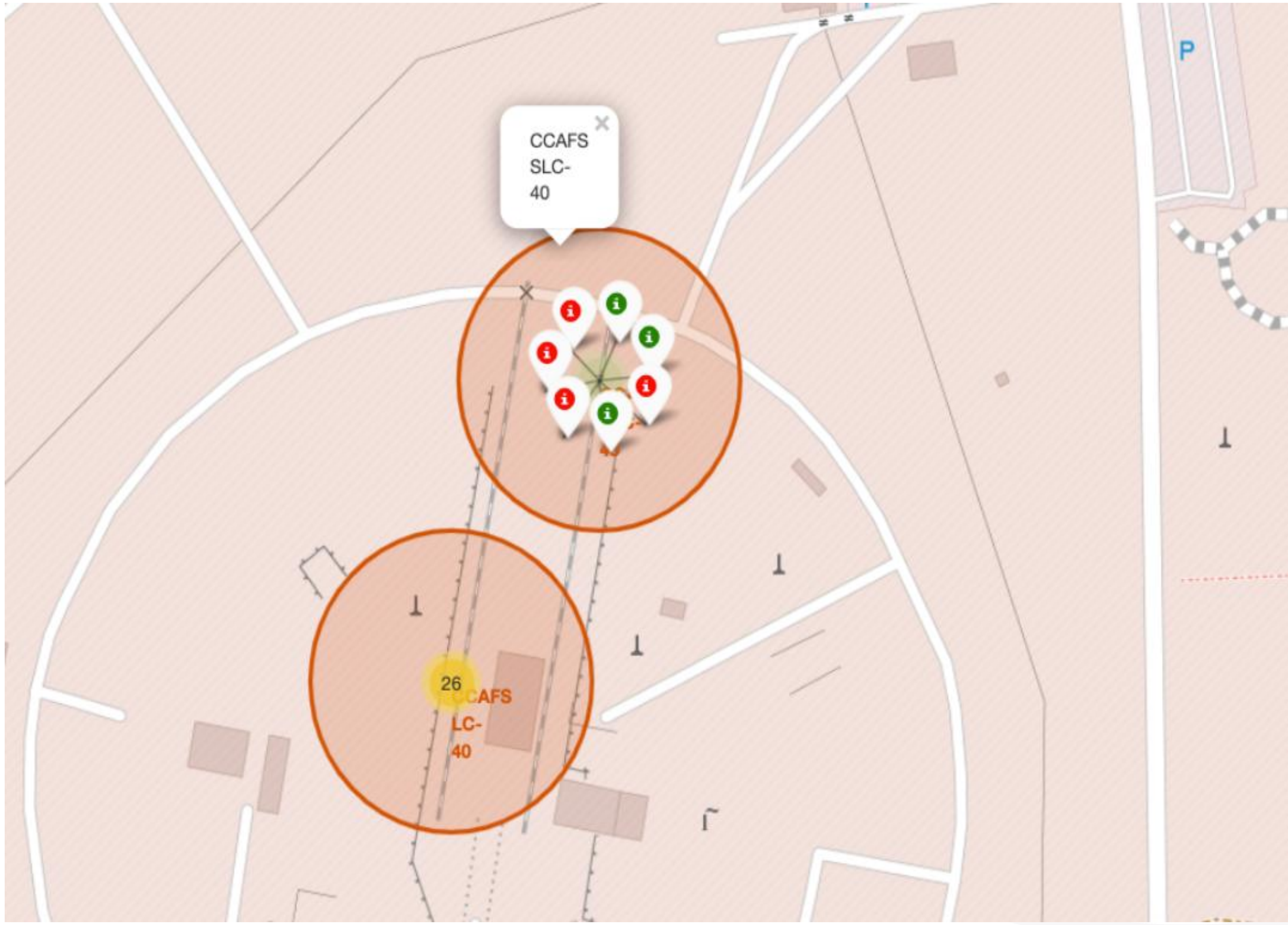
Interactive Launch Site Map (Folium)

Launch Site Distribution (Folium)



- **Key Findings:**
- Visualized Falcon 9 launch sites using Folium.
- Interactive markers display launch site locations and launch counts.
- CCAFS SLC-40 recorded the highest number of Falcon 9 launches, indicating it was the most active launch site.
- Geographic analysis helped compare launch site distribution.
- Location-based visualization helped identify launch site patterns and supported predictive analysis.

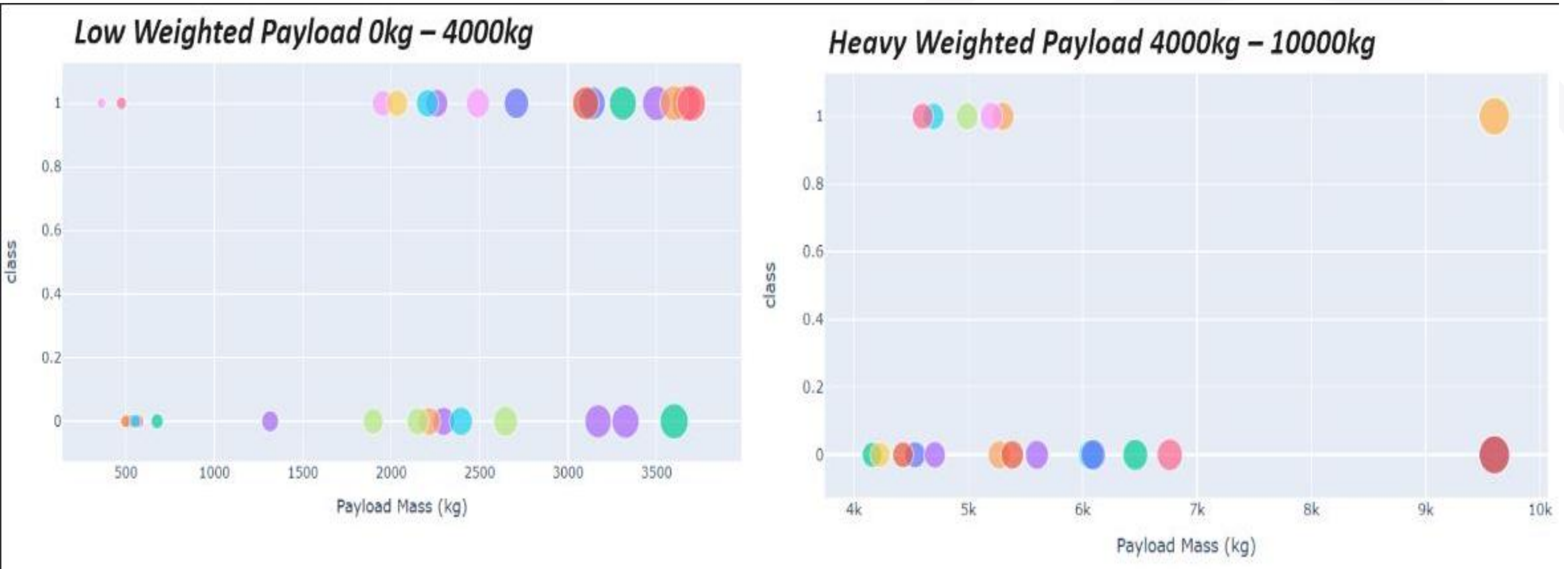
Interactive Marker Cluster Analysis



- **Key Findings:**
- Folium was used to visualize individual Falcon 9 launch records at **CCAFS SLC-40**.
- **Green markers** represent successful launches, while **red markers** represent failed launches.
- Marker clustering improves map readability when multiple launches occur at the same site.
- The visualization supports geographic analysis and helps compare launch outcomes across the launch site.

Payload Analysis

Payload vs Launch Success



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

- **Key Findings:**
- Payload mass influences launch success differently across booster versions.
- Lower payloads generally show higher success rates.
- Scatter plot helps identify relationships between payload, booster version, and landing outcome.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Conclusion

- **Key Takeaways**
- The analysis identified several factors that influence SpaceX Falcon 9 landing success.
- Flight Number showed a positive relationship with landing success.
- Payload mass, orbit type, and launch site were key factors influencing landing success.
- The project successfully combined **SQL, Python, Plotly Dash, and Folium** to analyze Falcon 9 launch data and support predictive insights.
- **Future Work:** Incorporate weather conditions and real-time launch data to further improve prediction accuracy.

- **Overall Findings**
- Landing success rates improved significantly over time, reaching their highest levels by 2019 and remaining consistently high thereafter.
- Historical launch data provides valuable insights into Falcon 9 landing success and supports data-driven decision-making.
- The Random Forest classifier achieved the best prediction performance among the evaluated machine learning models.
- Combining SQL analysis, EDA, Folium mapping, Plotly Dash, and machine learning provided a comprehensive understanding of Falcon 9 landing success.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

GITHUB REPOSITORY

GitHub Repository

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

APPENDIX

- **Additional Analysis**
- SQL query outputs used for exploratory analysis.
- Interactive Plotly Dash dashboard for exploratory data analysis.
- Folium map showing launch site proximity.
- Additional EDA charts supporting model development.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

Thank You!

Questions?